

SOP-03: Chloroform and Chloroform Mixtures (PHS)

Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
Chloroform	67-66-3
Chloroform, stabilized with ethanol	67-66-3
Chloroform : isoamyl alcohol mixture (24:1)	67-66-3 / 123-51-3

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Health hazard (carcinogen, STOT); Exclamation mark (acute toxicity Cat 4, irritant); Environment (aquatic toxicity)

- PARTICULARLY HAZARDOUS SUBSTANCE. Chloroform is classified as a select carcinogen (GHS Carcinogenicity Category 2; IARC Group 2B; NTP Reasonably Anticipated to be a Human Carcinogen) and a suspected reproductive toxicant (GHS Reproductive toxicity Category 2). Sections 9-11 of this SOP are mandatory.
- Stock chemicals: chloroform (2 L, J.T. Baker), chloroform stabilized with ethanol (1 L), and chloroform:isoamyl alcohol 24:1 (100 mL). Intermediates: phenol:chloroform:isoamyl alcohol extraction mixtures prepared or used during nucleic acid purification; chloroform added to TRIzol/TRI Reagent lysates for phase separation. Wastes: chloroform-containing organic phases, chloroform-contaminated aqueous phases, and chloroform-contaminated solid debris. No catalysts or inert compounds are used.

- Routes of exposure: inhalation of vapor (primary), skin absorption, eye contact, ingestion.
- How exposure might occur: opening and pouring chloroform on the open bench (prohibited by this SOP), adding chloroform to lysate tubes, vortexing and inverting tubes, opening tubes after centrifugation when the headspace is saturated with chloroform vapor, pipetting the aqueous phase off a chloroform layer, and disposal of organic phase to waste.
- Target organs: liver, kidney, central nervous system, heart (sensitization of the myocardium to catecholamines), skin, eyes, and the developing fetus.
- Signs and symptoms of exposure: dizziness, headache, fatigue, drowsiness, nausea and vomiting; at high concentrations, loss of consciousness, cardiac arrhythmia and death. Repeated skin contact causes dryness and dermatitis. Chronic exposure is associated with liver and kidney injury and with cancer of the liver and kidney in animal studies.
- Expected by-products: chloroform can slowly oxidize in air and light to phosgene, a highly toxic gas, especially in unstabilized product; this is why the ethanol-stabilized product is preferred and why containers are kept tightly closed, in the dark, and are not used past their expiration date. Chloroform can also form phosgene on contact with hot surfaces or open flames.
- GHS classification (from MyChem / SDS): Acute toxicity Category 3 or 4 (inhalation); Acute toxicity Category 4 (oral); Carcinogenicity Category 2; Reproductive toxicity Category 2; Serious eye damage/eye irritation Category 1 or 2A; Skin corrosion/irritation Category 2; Specific target organ toxicity - repeated exposure Category 1 or 2; Specific target organ toxicity - single exposure Category 3 (drowsiness/dizziness); Acute and chronic aquatic toxicity Category 3.
- Regulatory information: chloroform is a WISHA/OSHA-regulated substance with a ceiling PEL of 50 ppm (WAC 296-841 / 29 CFR 1910.1000 Table Z-2); NIOSH REL is 2 ppm as a 60-minute ceiling and NIOSH designates chloroform a potential occupational carcinogen. ACGIH TLV is 10 ppm (8-hr TWA), A3. Chloroform is a listed hazardous air pollutant and a listed hazardous waste constituent.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- ALL work with chloroform and chloroform mixtures - without exception - is performed inside the certified chemical fume hood in FTR 209 with the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S; verify the current EH&S certification sticker and confirm airflow (flow indicator or tissue check) before starting.
- Tubes containing chloroform are opened only inside the fume hood. This includes opening tubes after centrifugation - the headspace is saturated with chloroform vapor and venting it at the bench is a significant exposure.
- Chloroform is transported between the fume hood and the microcentrifuge in capped tubes held in a closed, leak-proof secondary container.
- Chloroform is never used near an open flame, hot plate, or hot surface (phosgene formation).
- A disposable absorbent bench liner is placed on the fume hood floor for all chloroform work and is discarded as hazardous waste at the end of the session.
- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly by lab staff. A chlorinated-solvent-compatible spill kit is maintained in the room.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.

- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: standard barrier lab coat, buttoned. Because chloroform is toxic by skin contact and absorption, additional protective clothing is required where splashes or skin contact is foreseeable: disposable chemically resistant sleeves for any transfer of more than 10 mL, and a chemically resistant apron plus face shield for any transfer of more than 100 mL or for spill cleanup.
- Contaminated lab coats are removed immediately, bagged, and laundered through the SAFS lab coat service or discarded as hazardous waste if saturated; they are never taken home.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Standard thin nitrile gloves offer only seconds to a few minutes of protection against chloroform and are NOT adequate as the sole barrier. Required: double-gloving with an inner 4-6 mil nitrile glove and an outer glove of a chloroform-resistant material - laminate film (North Silver Shield / Ansell AlphaTec 02-100) or Viton for any transfer of more than 10 mL or for any task with foreseeable contact.
- For brief, incidental-contact microliter pipetting inside the fume hood, double 6-8 mil nitrile gloves are acceptable PROVIDED the outer glove is removed and replaced immediately on any contact and at least every 15 minutes of active handling.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with chloroform. Gloves must be inspected prior to use, including a check for pinholes.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with chloroform. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for all chloroform work, including microliter-scale pipetting. A face shield worn over goggles is required for transfers of more than 100 mL and for spill cleanup.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, because all work is confined to a certified chemical fume hood.
- Chloroform is NEVER permitted to be used outside of the chemical fume hood under this SOP. There is no approved open-bench chloroform operation in this laboratory. If an operation cannot be performed in the fume hood, work must stop and the PI must contact EH&S.
- No lab member is currently enrolled in the EH&S Respiratory Protection Program for chloroform. Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit or Occupational Exposure Limit has been exceeded or when there is a possibility that an AL/OEL will be exceeded.
- If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed.

Program enrollment includes medical evaluation, training and fit testing. Where air-purifying respirators are appropriate, use a full-face respirator with organic vapor cartridges as a backup to engineering controls; note that organic vapor cartridges have poor service life against chloroform. Use a full-face supplied air respirator if it is the sole means of protection.

Section 4 - Special Handling and Storage Requirements

- Quantity limits: no more than 50 mL of chloroform may be dispensed into working containers at one time. Routine extraction work uses 100-500 microliters per sample. Procedures requiring more than 100 mL of chloroform in a single operation require prior written PI approval and a documented hazard review.
- Chloroform work is performed only during normal business hours when at least one other person is present in or immediately adjacent to FTR 209. No chloroform work is performed alone after hours.
- Unattended operations with chloroform are prohibited.
- A 'CHLOROFORM IN USE - PARTICULARLY HAZARDOUS SUBSTANCE' sign is posted on the fume hood sash while work is in progress and is removed when the hood has been decontaminated.
- Do not use chloroform with, or store it near, strong bases (sodium hydroxide - risk of violent reaction and dichlorocarbene formation), strong oxidizers, sodium or potassium metal, or acetone in the presence of base.
- Storage: chloroform is stored in the amber glass supplier bottle inside a labeled, chemically resistant secondary containment tray in the dedicated solvent cabinet in FTR 209, away from light and heat, tightly closed, and segregated from bases and oxidizers. Check the expiration/opened date; do not use product opened more than 12 months previously without checking for phosgene test strip results or replacing it. Chloroform is not stored in a household refrigerator.
- Transport: chloroform bottles are carried in a rubber or polyethylene bottle carrier and are not transported outside FTR 209.
- Best practices to minimize accidents: pre-stage all tubes, pipettes and waste containers inside the hood before opening the chloroform bottle; work over an absorbent liner; keep the waste container inside the hood; never carry an open chloroform container.
- Clean the chemical fume hood upon completion of tasks with a detergent solution followed by 70% ethanol and deionized water; wipe dry.
- Clean all contaminated surfaces with detergent solution followed by 70% ethanol and dry.
- Place all contaminated disposable items in appropriate laboratory waste for disposal.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with chloroform must be decontaminated at the end of the laboratory work session. Complete this inside the chemical fume hood before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- Small spill - less than 25 mL contained entirely INSIDE the operating fume hood: may be handled by trained lab staff. Leave the hood running with the sash low. Wearing goggles, face shield, laminate gloves over nitrile and a lab coat

with disposable sleeves, cover the spill with the chlorinated-solvent absorbent pads or vermiculite from the spill kit, work from the outside inward, and place used absorbent directly into a screw-top waste container inside the hood.

- Any spill greater than 25 mL, ANY spill outside the fume hood regardless of volume, or any spill in which a person may have been exposed: DO NOT clean up. Evacuate FTR 209 immediately, close the door, post a warning sign, prevent entry, and call 9-1-1 from a campus phone. Notify EH&S at 206-543-0467 during business hours or via 206-685-UWPD (8973) after hours.
- Chloroform is a dense, volatile liquid; even a small spill outside a hood can produce airborne concentrations above the ceiling PEL in a short time in a small room. Err on the side of evacuation.
- Liquid spills only are anticipated; chloroform is not present in solid form.
- PPE required for cleanup: chemical splash goggles, face shield, laminate or Viton gloves over 6-8 mil nitrile, buttoned lab coat with disposable chemically resistant sleeves and apron, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - CHEMICAL SPILL' on the door for any spill outside the hood; restrict entry until EH&S clears the space.
- Spill area must be cleaned up in the following manner: after absorbent removal, clean the spill area thoroughly with a detergent solution followed by clean water, then wipe with 70% ethanol and dry. Repeat the detergent wash a second time for chloroform.
- Spill cleanup materials must be disposed of in the following manner: double bag all absorbent and contaminated disposables, place in a screw-top container inside the fume hood, label with a UW Hazardous Waste Label listing 'chloroform - spill debris', and submit a collection request to EH&S promptly rather than accumulating it.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Inhalation exposure: move the person out of the contaminated area to fresh air immediately. Call 9-1-1. Chloroform sensitizes the heart to adrenaline - keep the person calm and still and do not administer adrenaline-type medications unless directed by a physician. If breathing has stopped, trained personnel should give rescue breathing; rescuers must not enter an area with high chloroform vapor without appropriate protection.
- Skin exposure: use the nearest safety shower for 15 minutes; stay under the shower and remove all contaminated clothing including shoes; use a clean lab coat or spare clothing for cover-up.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open. Seek medical evaluation - chloroform can cause corneal injury.
- Ingestion: do not induce vomiting (aspiration hazard); rinse mouth with water and call 9-1-1.
- Sharps injury (needle stick or subcutaneous exposure): scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap; seek medical evaluation.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent (chloroform), dose, route of exposure, and time since exposure. State that chloroform is a suspect carcinogen and a cardiac sensitizer. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

- All chloroform exposures are reported to the PI and to EH&S even if the person feels well; EH&S will determine whether medical surveillance is warranted.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) neat chloroform and chloroform:isoamyl alcohol; (b) chloroform-containing organic phase and interphase from nucleic acid extractions, which typically also contains phenol and guanidinium thiocyanate; (c) aqueous phases contaminated with residual chloroform; (d) chloroform-contaminated solid debris (tips, tubes, liners, absorbent, gloves).
- Keep the halogenated (chlorinated) organic waste stream SEPARATE from the non-halogenated solvent waste stream in SOP-01. Mixing them substantially increases disposal cost and can create incompatible mixtures. Label the container 'HALOGENATED ORGANIC WASTE'.
- NEVER add chloroform waste to a container holding sodium hydroxide or any strong base, strong oxidizer, sodium metal, or acetone-plus-base.
- Accumulate liquid waste at the point of generation - inside the fume hood - in a sturdy, chemically compatible container (glass or fluorinated HDPE; standard polyethylene is not suitable for long-term chloroform storage) with a securely closable screw-top lid. Keep the container closed except when actively adding waste, and keep it in secondary containment.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage (e.g., 'chloroform 60%, phenol 30%, guanidinium thiocyanate 10%'), starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Do NOT pour any chloroform-containing liquid down the drain. Do NOT evaporate chloroform in the fume hood as a disposal method. No neutralization or in-lab treatment of chloroform waste is permitted.
- Solid debris contaminated with chloroform is collected in a lined, closed, labeled solid hazardous waste container kept in or immediately adjacent to the fume hood - never in the regular trash.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions. Request pickup promptly; do not let chloroform waste accumulate beyond the satellite accumulation area limits.
- Update the MyChem inventory when a chloroform container is emptied, received, or transferred.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials.

Section 7 - Protocol

- Procedure A - Phase separation of TRIzol / TRI Reagent / RNAzol lysates. Inside the certified fume hood, on an absorbent liner: add 0.2 mL chloroform per 1 mL of lysate to the capped 1.5-2 mL tube. Cap securely, shake vigorously by hand for 15 seconds (do not vortex open), and incubate 2-3 minutes at room temperature inside the hood. Transfer the closed tubes in a covered secondary container to the microcentrifuge; centrifuge 12,000 x g, 15 min, 4 degC. Return the closed tubes to the fume hood and open them there. Pipette the upper aqueous phase into a clean tube; transfer the interphase and organic phase directly into the halogenated organic waste container inside the hood.

- Procedure B - Phenol:chloroform:isoamyl alcohol extraction. Inside the fume hood, add an equal volume of 25:24:1 phenol:chloroform:isoamyl alcohol (or 24:1 chloroform:isoamyl alcohol) to the capped aqueous sample. Mix by inversion, centrifuge as above, and recover the aqueous phase inside the hood. See SOP-05 for the phenol-specific requirements that apply concurrently.
- Procedure C - Dispensing from the stock bottle. Inside the fume hood, place the stock bottle on the absorbent liner, uncap, dispense into a labeled glass or fluoropolymer container using a glass or polypropylene serological pipette (chloroform attacks some plastics and dissolves marker ink), recap the stock bottle immediately, and return it to the cabinet before proceeding.
- Prohibited: heating chloroform, distilling chloroform, using chloroform with strong base, using chloroform outside the fume hood, and use of chloroform by untrained or unauthorized personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

No. N/A - chloroform is not administered to animals under this SOP and is not used for euthanasia or anesthesia in this laboratory. Chloroform must never be used as an animal anesthetic.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 9 - Approvals Required

- All staff working with chloroform must be trained on this SOP prior to starting work. They must also review the chloroform SDS, and it must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.
- Chloroform is a particularly hazardous substance. Written authorization from the Principal Investigator is required before any individual may work with chloroform, and that authorization is recorded on the training roster in Section 12 with the PI's countersignature.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the chloroform SDS; (4) documented instruction on fume hood use and airflow verification.
- A worker must complete at least two supervised runs of Procedure A or B, observed and signed off by the PI or a designated experienced lab member, before performing chloroform work independently for the first time.
- Personnel who are pregnant, planning a pregnancy, or breastfeeding should consult the EH&S Reproductive Health in the Workplace resources and their healthcare provider, and are encouraged to discuss alternative task assignment with the PI, because chloroform is a suspected reproductive toxicant. Such consultation is voluntary and confidential; no disclosure to the PI is required.
- No medical examination is required for the work described because respirators are not used. A medical examination and fit testing through the EH&S Respiratory Protection Program would be required before any respirator use.

Section 10 - Decontamination

- Fume hood: at the end of every chloroform session, remove and bag the absorbent bench liner as hazardous solid waste, then wipe the entire hood work surface and the inside of the sash with a detergent solution, followed by clean water, followed by 70% ethanol; allow to dry with the hood running.
- Equipment: wipe pipettors, tube racks and the outside of secondary containers with detergent solution then 70% ethanol. Centrifuge rotors and buckets used for chloroform tubes are wiped with detergent solution and 70% ethanol after each run and inspected for residue; a cracked or leaking tube requires the rotor and chamber to be decontaminated inside the hood or with the centrifuge lid closed and the room ventilated.
- Glassware: rinse three times inside the fume hood, collecting all rinses in the halogenated organic waste container for the first rinse and the aqueous waste container for subsequent rinses; then wash with detergent and water and rinse three times with deionized water.
- Controlled areas: there are no glove boxes or restricted-access hoods in FTR 209. The chloroform designated area (Section 11) is wiped down as described above at the end of each session and monthly regardless of use.
- Cleaning solutions and materials: laboratory detergent (e.g., Alconox or Contrad) solution, deionized water, 70% ethanol, lint-free wipes, and disposable absorbent bench liners.

Section 11 - Designated Area

- DESIGNATED AREA: the certified chemical fume hood in FTR 209 is the sole designated area for all use of chloroform and chloroform mixtures.
- The designated area is posted with a 'DESIGNATED AREA - PARTICULARLY HAZARDOUS SUBSTANCE - CHLOROFORM' sign listing the PI, contact number and the requirement for authorization and PPE.
- Chloroform-containing waste containers are also kept within the designated area, in secondary containment, inside the fume hood.
- Chloroform stock storage: the labeled solvent cabinet in FTR 209 with secondary containment; the cabinet is posted with the same PHS designation.
- No other bench, room or piece of equipment in the Roberts Lab space is authorized for open chloroform handling.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.
- Because this SOP covers a particularly hazardous substance, the Principal Investigator countersigns each entry below to record the written authorization required by Section 9.

I have read and understand the content of this SOP:

Name (print)	Signature	Date

[illegible]